JAVA ASSIGNMENT 2

The Employee Assignment

Create a menu based console application for holding the data of the Employees in an Organisation. The details of the Employee entered by the user from the console will be stored in a doubly linked list which can be then traversed by the user by choosing the appropriate menu selection. Each sub menu of the section will loop until the Exit to Main Menu option is not selected.

1. Add an Employee
a. Manager
b. Engineer
c. Sales Person
d. Exit to Main Menu
2. Display
a. All Employees
b. First Employee
c. Next Employee
d. Previous Employee
e. Last Employee
f. Exit to Main Menu
3. Sort
a. All Managers
b. All Engineers
c. All Sales Person
d. All Employees Alphabetic order ascending
e. All Employees Alphabetic order descending
f. Exit to Main Menu
4. Save to File
5. Load from File
6. Exit

CODE:

```
import java.util.Scanner;
public class Employee2 {
    static int id[] = new int[100];
    static String name[] = new String[100];
    static double salary[] = new double[100];
    static String type[] = new String[100];
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int count = 0;
        int current = 0;
        int choice;
        do {
            System.out.println();
            System.out.println("----- EMPLOYEE MENU -----");
            System.out.println("1. Add an Employee");
            System.out.println("2. Display");
            System.out.println("3. Sort");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            if(choice == 1) {
                int addChoice;
                do {
                    System.out.println();
                    System.out.println("----- ADD EMPLOYEE -----");
                    System.out.println("1. Manager");
                    System.out.println("2. Engineer");
                    System.out.println("3. Sales Person");
                    System.out.println("4. Exit to Main Menu");
                    System.out.print("Enter your choice: ");
                    addChoice = sc.nextInt();
                    if(addChoice >= 1 && addChoice <= 3) {
                        System.out.print("Enter Employee ID: ");
                        id[count] = sc.nextInt();
                        sc.nextLine();
```

```
            System.out.print("Enter Employee Name: ");
            name[count] = sc.nextLine();
            System.out.print("Enter Salary: ");
            salary[count] = sc.nextDouble();
            if(addChoice == 1) {
               type[count] = "Manager";
            } else if(addChoice == 2) {
               type[count] = "Engineer";
            } else {
               type[count] = "Sales Person";
            }
            count++;
            System.out.println("Employee added successfully.");
         }
      } while(addChoice != 4);
   }
   else if(choice == 2) {
      if(count == 0) {
         System.out.println("No employees available.");
      } else {
         int displayChoice;
         do {
            System.out.println();
            System.out.println("----- DISPLAY MENU -----");
            System.out.println("1. All Employees");
            System.out.println("2. First Employee");
            System.out.println("3. Next Employee");
            System.out.println("4. Previous Employee");
            System.out.println("5. Last Employee");
            System.out.println("6. Exit to Main Menu");
            System.out.print("Enter your choice: ");
            displayChoice = sc.nextInt();
            if(displayChoice == 1) {
               for(int i = 0; i < count; i++) {
                  System.out.println();
```

```
            System.out.println("ID: " + id[i]);
            System.out.println("Name: " + name[i]);
            System.out.println("Type: " + type[i]);
            System.out.println("Salary: " + salary[i]);
        }
    } else if(displayChoice == 2) {
        current = 0;
        System.out.println();
        System.out.println("ID: " + id[current]);
        System.out.println("Name: " + name[current]);
        System.out.println("Type: " + type[current]);
        System.out.println("Salary: " + salary[current]);
    } else if(displayChoice == 3) {
        if(current < count - 1) {
            current++;
            System.out.println();
            System.out.println("ID: " + id[current]);
            System.out.println("Name: " + name[current]);
            System.out.println("Type: " + type[current]);
            System.out.println("Salary: " + salary[current]);
        } else {
            System.out.println("Already at last employee.");
        }
    } else if(displayChoice == 4) {
        if(current > 0) {
            current--;
            System.out.println();
            System.out.println("ID: " + id[current]);
            System.out.println("Name: " + name[current]);
            System.out.println("Type: " + type[current]);
            System.out.println("Salary: " + salary[current]);
        } else {
            System.out.println("Already at first employee.");
        }
    } else if(displayChoice == 5) {
```

```
                    current = count - 1;
                    System.out.println();
                    System.out.println("ID: " + id[current]);
                    System.out.println("Name: " + name[current]);
                    System.out.println("Type: " + type[current]);
                    System.out.println("Salary: " + salary[current]);
                }
            } while(displayChoice != 6);
        }
    }
    else if(choice == 3) {
        if(count == 0) {
            System.out.println("No employees available.");
        } else {
            int sortChoice;
            do {
                System.out.println();
                System.out.println("----- SORT MENU -----");
                System.out.println("1. All Managers");
                System.out.println("2. All Engineers");
                System.out.println("3. All Sales Person");
                System.out.println("4. All Employees Alphabetic
Ascending");
                System.out.println("5. All Employees Alphabetic
Descending");
                System.out.println("6. Exit to Main Menu");
                System.out.print("Enter your choice: ");
                sortChoice = sc.nextInt();
                if(sortChoice == 1) {
                    System.out.println("Managers:");
                    for(int i = 0; i < count; i++) {
                        if(type[i].equals("Manager")) {
                            System.out.println(
                                    id[i] + " " +
                                            name[i] + " " +
```

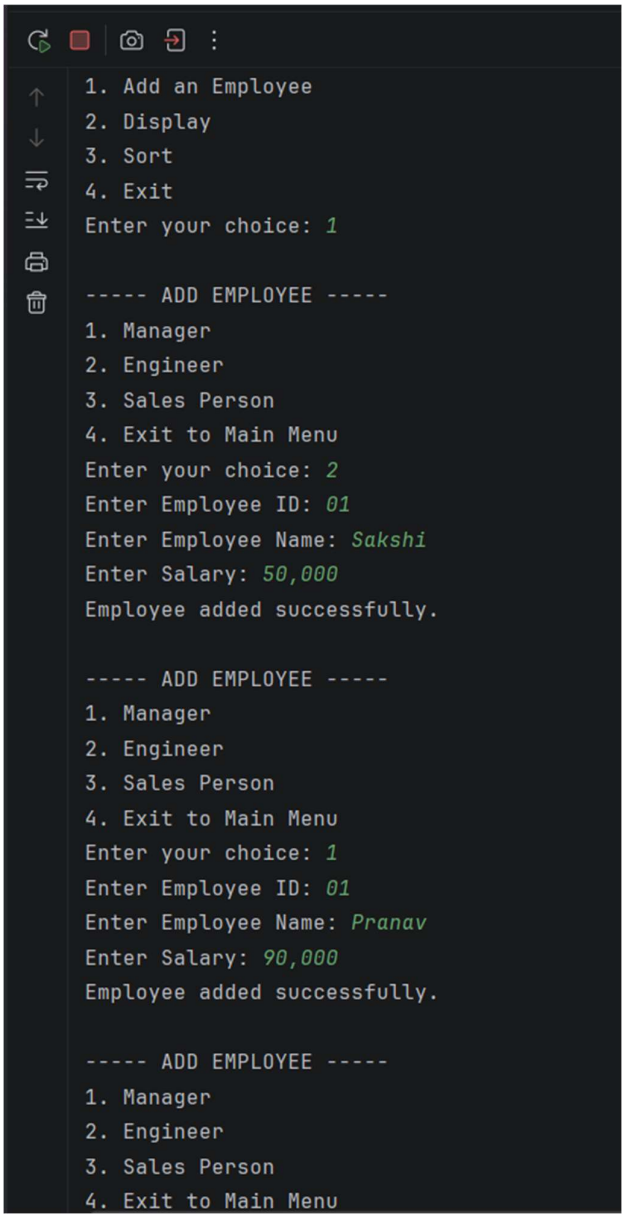
```
                    salary[i]
                );
            }
        }
    } else if(sortChoice == 2) {
        System.out.println("Engineers:");
        for(int i = 0; i < count; i++) {
            if(type[i].equals("Engineer")) {
                System.out.println(
                    id[i] + " " +
                        name[i] + " " +
                        salary[i]
                );
            }
        }
    } else if(sortChoice == 3) {
        System.out.println("Sales Persons:");
        for(int i = 0; i < count; i++) {
            if(type[i].equals("Sales Person")) {
                System.out.println(
                    id[i] + " " +
                        name[i] + " " +
                        salary[i]
                );
            }
        }
    } else if(sortChoice == 4) {
        for(int i = 0; i < count; i++) {
            for(int j = i + 1; j < count; j++) {
                if(name[i].compareToIgnoreCase(name[j]) > 0) {
                    String tempName = name[i];
                    name[i] = name[j];
                    name[j] = tempName;
                    int tempId = id[i];
                    id[i] = id[j];
```

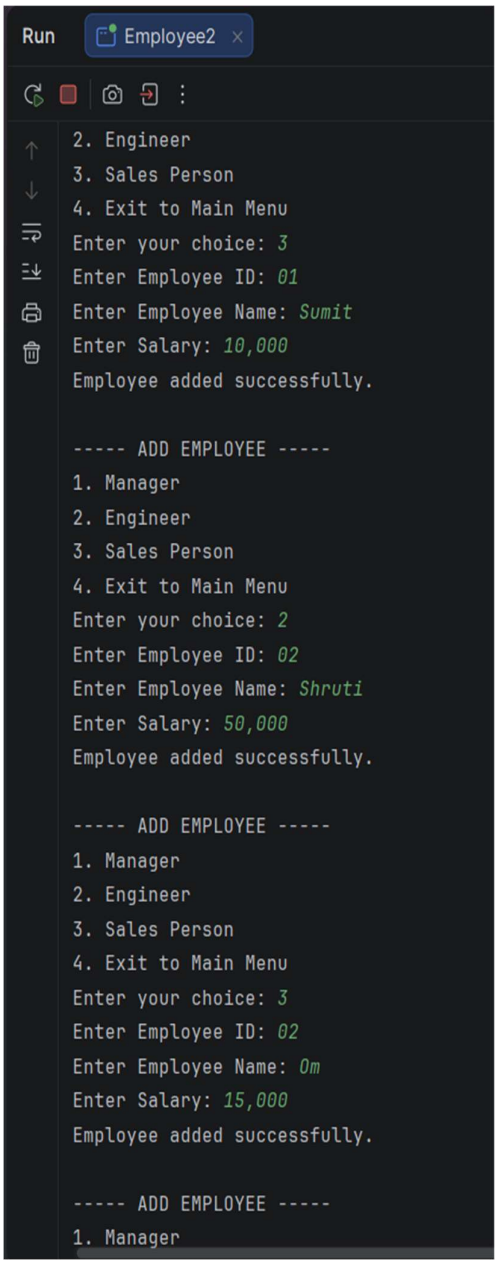
```
                id[j] = tempId;
                double tempSalary = salary[i];
                salary[i] = salary[j];
                salary[j] = tempSalary;
                String tempType = type[i];
                type[i] = type[j];
                type[j] = tempType;
            }
        }
    }
    System.out.println("Employees in Ascending Order:");
    for(int i = 0; i < count; i++) {
        System.out.println(name[i]);
    }
} else if(sortChoice == 5) {
    for(int i = 0; i < count; i++) {
        for(int j = i + 1; j < count; j++) {
            if(name[i].compareToIgnoreCase(name[j]) < 0) {
                String tempName = name[i];
                name[i] = name[j];
                name[j] = tempName;
                int tempId = id[i];
                id[i] = id[j];
                id[j] = tempId;
                double tempSalary = salary[i];
                salary[i] = salary[j];
                salary[j] = tempSalary;
                String tempType = type[i];
                type[i] = type[j];
                type[j] = tempType;
            }
        }
    }
    System.out.println("Employees in Descending Order:");
    for(int i = 0; i < count; i++) {
```

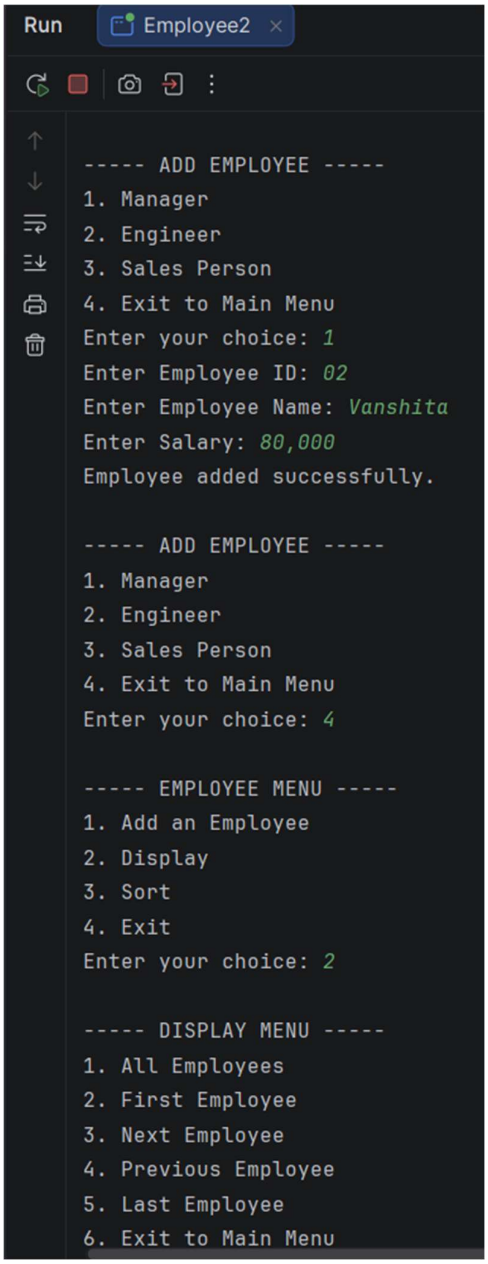
```
                    System.out.println(name[i]);
                }
            }
        } while(sortChoice != 6);
    }
}
else if(choice == 4) {
    System.out.println("Program exited.");
}
else {

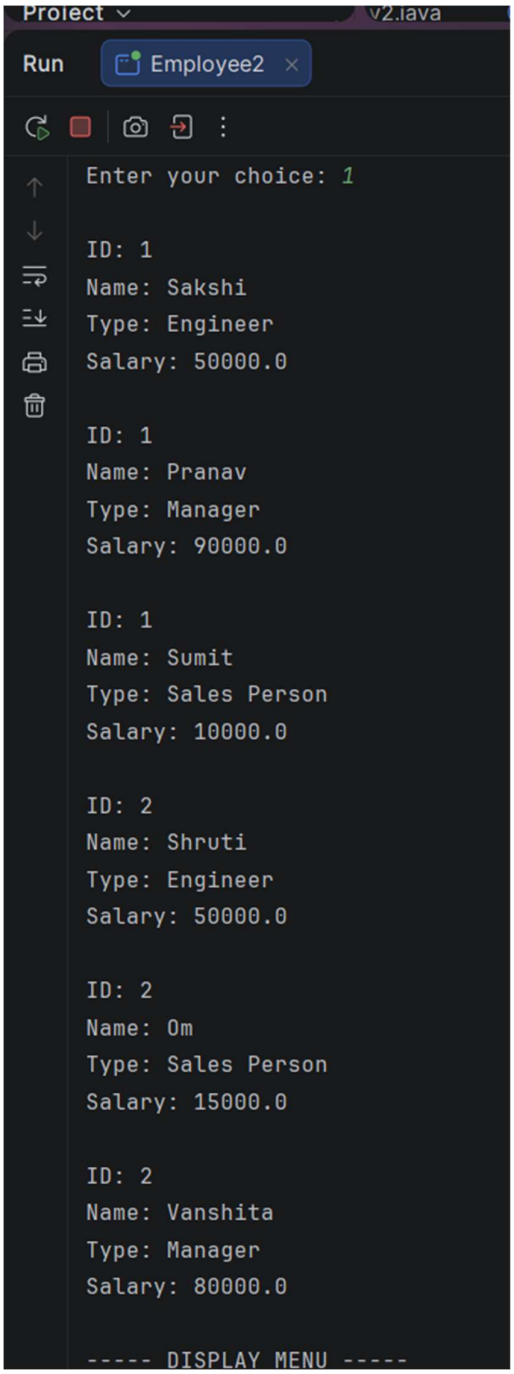
    System.out.println("Invalid choice.");
}
} while(choice != 4);
sc.close();
}
}
```

Output:


1. Add an Employee
2. Display
3. Sort
4. Exit
Enter your choice: 1
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 2
Enter Employee ID: 01
Enter Employee Name: Sakshi
Enter Salary: 50,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 1
Enter Employee ID: 01
Enter Employee Name: Pranav
Enter Salary: 90,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu


Run
Employee2
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 3
Enter Employee ID: 01
Enter Employee Name: Sumit
Enter Salary: 10,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 2
Enter Employee ID: 02
Enter Employee Name: Shruti
Enter Salary: 50,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 3
Enter Employee ID: 02
Enter Employee Name: Om
Enter Salary: 15,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager


Run
Employee2
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 1
Enter Employee ID: 02
Enter Employee Name: Vanshita
Enter Salary: 80,000
Employee added successfully.
----- ADD EMPLOYEE -----
1. Manager
2. Engineer
3. Sales Person
4. Exit to Main Menu
Enter your choice: 4
----- EMPLOYEE MENU -----
1. Add an Employee
2. Display
3. Sort
4. Exit
Enter your choice: 2
----- DISPLAY MENU -----
1. All Employees
2. First Employee
3. Next Employee
4. Previous Employee
5. Last Employee
6. Exit to Main Menu


Project
v2.java
Run
Employee2
Enter your choice: 1
ID: 1
Name: Sakshi
Type: Engineer
Salary: 50000.0
ID: 1
Name: Pranav
Type: Manager
Salary: 90000.0
ID: 1
Name: Sumit
Type: Sales Person
Salary: 10000.0
ID: 2
Name: Shruti
Type: Engineer
Salary: 50000.0
ID: 2
Name: Om
Type: Sales Person
Salary: 15000.0
ID: 2
Name: Vanshita
Type: Manager
Salary: 80000.0
----- DISPLAY MENU -----


Salary: 80000.0
----- DISPLAY MENU -----
1. All Employees
2. First Employee
3. Next Employee
4. Previous Employee
5. Last Employee
6. Exit to Main Menu
Enter your choice: 6
----- EMPLOYEE MENU -----
1. Add an Employee
2. Display
3. Sort
4. Exit
Enter your choice: 3
----- SORT MENU -----
1. All Managers
2. All Engineers
3. All Sales Person
4. All Employees Alphabetic Ascending
5. All Employees Alphabetic Descending
6. Exit to Main Menu
Enter your choice: 1
Managers:
1 Pranav 90000.0
2 Vanshita 80000.0
----- SORT MENU -----
1. All Managers
2. All Engineers